

Burn Density Analysis by Local Government Area (LGA)

Objective

The objective of this project was to calculate and visualize bushfire burn density across Local Government Areas (LGAs) in South Australia. Burn density was calculated by dividing the total burned area within each LGA by the total area of that LGA.

Study Area

South Australia, Australia

Data Sources

- Fire History Dataset, Data SA (Government of South Australia)
 - Local Government Areas (LGA), Data SA (Government of South Australia)
 - National Parks, Data SA (Government of South Australia)
 - Esri World Topographic Basemap
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Software

- ArcGIS Pro 3.6
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Projection

GDA2020 / Australian Albers (EPSG:9473)

GIS Workflow

1. Buffered National Parks by 5 km.
2. Intersected Fire History with the buffered National Parks.

3. Projected the dataset to GDA2020 / Australian Albers.
4. Calculated burn area in square kilometres.
5. Summarized burn area within each Local Government Area using the **Summarize Within** tool.
6. Calculated Burn Density using the formula:

Burn Density = Total Burn Area (km²) / LGA Area (km²)

7. Created a graduated colour map to visualize burn density across South Australia.
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Key Findings

- Burn density was successfully calculated for all Local Government Areas.
 - Higher burn density was identified in parts of northern and southern South Australia.
 - The analysis provides a clear comparison of bushfire impact between LGAs.
 - The final map presents spatial patterns that can support regional fire management and planning.
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GIS Skills Demonstrated

- Data preparation
 - Spatial data management
 - Buffer Analysis
 - Intersect Analysis
 - Coordinate System Projection
 - Geometry Calculation
 - Summarize Within
 - Attribute Calculation
 - Spatial Analysis
 - Cartographic Design
 - Professional Map Layout
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GIS Portfolio Project

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